SG No. 205 T_{h}^{6} $Pa\bar{3}$ [cubic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `4a`, site symmetry: `.-3.`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$[0,\ 0,\ 0]$	`[1,5,9,-13,-17,-21]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X,\ X,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\ -X,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,6,11,-16,-18,-23]`

Table 2: Wyckoff bond: `12b@4a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-13]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,-14]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,-15]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,-16]`
5	$[Z,\ X,\ Y]$	$[0,\ 0,\ 0]$	`[5,-17]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[6,-18]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,-19]`
8	$[-Z,\ X,\ -Y]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[8,-20]`
9	$[Y,\ Z,\ X]$	$[0,\ 0,\ 0]$	`[9,-21]`
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,-22]`
11	$[Y,\ -Z,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[11,-23]`
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `4b`, site symmetry: `.-3.`

Table 3: Wyckoff bond: `4a@4b`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,5,9,-13,-17,-21]`
2	$[-X,\ -X,\ X]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,7,12,-14,-19,-24]`
3	$[-X,\ X,\ -X]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\ -X,\ -X]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,6,11,-16,-18,-23]`

Table 4: Wyckoff bond: `12b@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,-13]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-14]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,-15]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,-16]`
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,-17]`
6	$[Z,\ -X,\ -Y]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,-18]`

continued ...

Table 4

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$\left[0, \frac{1}{2}, 0\right]$	[7,-19]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	[8,-20]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,-21]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	[10,-22]
11	$[Y, -Z, -X]$	$\left[0, 0, \frac{1}{2}\right]$	[11,-23]
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{2}, 0\right]$	[12,-24]

* Wyckoff site: 8c, site symmetry: .3.

Table 5: Wyckoff bond: 8a@8c, bond symmetry: .3.

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9]
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[2,7,12]
3	$[-X, X, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[3,8,10]
4	$[X, -X, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[4,6,11]
5	$[-X, -X, -X]$	$[-x, -x, -x]$	[13,17,21]
6	$[X, X, -X]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	[16,18,23]

Table 6: Wyckoff bond: 24b@8c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[12]
13	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[13]
14	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	[14]
15	$[X, -Y, Z]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[15]
16	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	[16]
17	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[17]
18	$[-Z, X, Y]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	[18]
19	$[Z, X, -Y]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	[19]
20	$[Z, -X, Y]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[20]
21	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[21]
22	$[Y, -Z, X]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[22]

continued ...

Table 6

No.	vector	center	mapping
23	$[-Y,\, Z,\, X]$	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, x\right]$	[23]
24	$[Y,\, Z,\, -X]$	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}-x\right]$	[24]

* Wyckoff site: 24d, site symmetry: 1

Table 7: Wyckoff bond: 24a@24d, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	[1]
2	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[2]
3	$[-X,\, Y,\, -Z]$	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[3]
4	$[X,\, -Y,\, -Z]$	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4]
5	$[Z,\, X,\, Y]$	$[z,\, x,\, y]$	[5]
6	$[Z,\, -X,\, -Y]$	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[6]
7	$[-Z,\, -X,\, Y]$	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[7]
8	$[-Z,\, X,\, -Y]$	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[8]
9	$[Y,\, Z,\, X]$	$[y,\, z,\, x]$	[9]
10	$[-Y,\, Z,\, -X]$	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[10]
11	$[Y,\, -Z,\, -X]$	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[11]
12	$[-Y,\, -Z,\, X]$	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[12]
13	$[-X,\, -Y,\, -Z]$	$[-x,\, -y,\, -z]$	[13]
14	$[X,\, Y,\, -Z]$	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[14]
15	$[X,\, -Y,\, Z]$	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15]
16	$[-X,\, Y,\, Z]$	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[16]
17	$[-Z,\, -X,\, -Y]$	$[-z,\, -x,\, -y]$	[17]
18	$[-Z,\, X,\, Y]$	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$	[18]
19	$[Z,\, X,\, -Y]$	$\left[z+\frac{1}{2},\, x,\, \frac{1}{2}-y\right]$	[19]
20	$[Z,\, -X,\, Y]$	$\left[z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[20]
21	$[-Y,\, -Z,\, -X]$	$[-y,\, -z,\, -x]$	[21]
22	$[Y,\, -Z,\, X]$	$\left[y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[22]
23	$[-Y,\, Z,\, X]$	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$	[23]
24	$[Y,\, Z,\, -X]$	$\left[y+\frac{1}{2},\, z,\, \frac{1}{2}-x\right]$	[24]